

# MedTrack

## Quality Assurance (QA) Checklist

*Hospital Operations & Resource Utilization Dashboard | Final QA Document*

**Purpose:** This checklist verifies the correctness, consistency, functionality, usability, integration, and final delivery readiness of the MedTrack dashboard suite.

**Status Convention:** ☐ Pass ☐ Fail ☐ N/A

All checklist items are marked as Pass in this finalized QA version.

### 1. Data Quality Validation

| ID    | QA Check                                   | Expected Result                                                | Status                                                                    |
|-------|--------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------|
| DQ-01 | Verify all required datasets are available | All required datasets are present                              | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DQ-02 | Verify dataset column names                | Required columns are present and correctly named               | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DQ-03 | Check missing values                       | Missing values are within the defined project threshold        | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DQ-04 | Check duplicate records                    | Unintended duplicate records are removed                       | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DQ-05 | Verify data types                          | Dates, numerical and categorical fields have appropriate types | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DQ-06 | Check department/service names             | Department names are standardized                              | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DQ-07 | Check whitespace                           | Leading/trailing spaces are removed                            | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DQ-08 | Validate numerical fields                  | No unexpected or invalid numerical values                      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DQ-09 | Validate date fields                       | Dates are correctly formatted and logically consistent         | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DQ-10 | Verify Tableau-ready dataset               | Final dataset contains required analytical fields              | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

### 2. KPI Validation

| ID     | QA Check              | Expected Result                          | Status                                                                    |
|--------|-----------------------|------------------------------------------|---------------------------------------------------------------------------|
| KPI-01 | Total Patients        | Value matches validated patient count    | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| KPI-02 | Average LOS           | Tableau value matches source calculation | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| KPI-03 | Occupancy Rate        | Values match the project definition      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| KPI-04 | Bed Utilization Rate  | Values match the project definition      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| KPI-05 | Readmission Rate      | Value matches validated calculation      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| KPI-06 | Department Efficiency | Value matches KPI-generation output      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

|        |                |                                         |                                                                           |
|--------|----------------|-----------------------------------------|---------------------------------------------------------------------------|
| KPI-07 | KPI formatting | Consistent formatting across dashboards | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| KPI-08 | KPI accuracy   | Accuracy meets project target           | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

### 3. Dashboard 1: Hospital Overview

| ID    | QA Check                            | Expected Result                    | Status                                                                    |
|-------|-------------------------------------|------------------------------------|---------------------------------------------------------------------------|
| HO-01 | KPI cards display correctly         | All required KPIs are visible      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| HO-02 | Admission share visualization       | Values and labels are correct      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| HO-03 | Department efficiency visualization | Departments and values are correct | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| HO-04 | Readmission visualization           | Values correspond to source data   | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| HO-05 | LOS visualization                   | Department LOS values are correct  | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| HO-06 | Occupancy visualization             | Values are correct                 | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| HO-07 | Bed utilization visualization       | Values are correct                 | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| HO-08 | Service/department filter           | Relevant visuals respond correctly | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| HO-09 | Tooltips                            | Correct details appear on hover    | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| HO-10 | Layout                              | No overlapping or cut-off objects  | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

### 4. Dashboard 2: Patient Flow

| ID    | QA Check               | Expected Result                     | Status                                                                    |
|-------|------------------------|-------------------------------------|---------------------------------------------------------------------------|
| PF-01 | Admission trend        | Monthly values are correct          | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| PF-02 | Discharge trend        | Monthly values are correct          | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| PF-03 | Patients by department | Department totals are correct       | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| PF-04 | LOS heatmap            | Department/month values are correct | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| PF-05 | Admitted vs refused    | Values match source data            | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| PF-06 | Department filter      | Correct worksheets respond          | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| PF-07 | Tooltips               | Correct patient-flow details appear | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| PF-08 | Visual readability     | Labels and values are readable      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

### 5. Dashboard 3: Department Analytics

| ID    | QA Check                  | Expected Result  | Status                                                                    |
|-------|---------------------------|------------------|---------------------------------------------------------------------------|
| DA-01 | Service Count KPI         | Value is correct | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DA-02 | Department Efficiency KPI | Value is correct | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DA-03 | Average Occupancy KPI     | Value is correct | <input checked="" type="checkbox"/> Pass                                  |

|       |                               |                                         |                                                                           |
|-------|-------------------------------|-----------------------------------------|---------------------------------------------------------------------------|
|       |                               |                                         | <input type="checkbox"/> Fail                                             |
| DA-04 | Efficiency visualization      | Department comparison is accurate       | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DA-05 | Occupancy visualization       | Values and target/reference are correct | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DA-06 | Bed utilization visualization | Department values are correct           | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DA-07 | Readmission visualization     | Values are correct                      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| DA-08 | Department filtering          | Relevant views update correctly         | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

## 6. Dashboard 4: Resource Analysis

| ID    | QA Check                    | Expected Result                         | Status                                                                    |
|-------|-----------------------------|-----------------------------------------|---------------------------------------------------------------------------|
| RA-01 | Average Bed Utilization KPI | Value is correct                        | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| RA-02 | Total Beds KPI              | Value is correct                        | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| RA-03 | Average Occupancy KPI       | Value is correct                        | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| RA-04 | Bed utilization comparison  | Department values are correct           | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| RA-05 | Occupancy comparison        | Department values are correct           | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| RA-06 | Resource summary table      | Values match source data                | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| RA-07 | Occupancy vs efficiency     | Both measures are correctly represented | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| RA-08 | Table headings              | No truncated or unclear headings        | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| RA-09 | Tooltips                    | Correct resource details appear         | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

## 7. Dashboard Integration

| ID     | QA Check                                 | Expected Result                              | Status                                                                    |
|--------|------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------|
| INT-01 | Hospital Overview → Patient Flow         | Correct dashboard opens                      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| INT-02 | Hospital Overview → Department Analytics | Correct dashboard opens                      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| INT-03 | Hospital Overview → Resource Analysis    | Correct dashboard opens                      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| INT-04 | Patient Flow → Hospital Overview         | Correct dashboard opens                      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| INT-05 | Patient Flow → Department Analytics      | Correct dashboard opens                      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| INT-06 | Patient Flow → Resource Analysis         | Correct dashboard opens                      | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| INT-07 | Department Analytics → other dashboards  | Navigation works correctly                   | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| INT-08 | Resource Analysis → other dashboards     | Navigation works correctly                   | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| INT-09 | Home button                              | Returns to Hospital Overview                 | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| INT-10 | Navigation consistency                   | Same navigation bar/design across dashboards | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

## 8. Visual and Usability QA

| ID    | QA Check             | Expected Result                             | Status                                                                    |
|-------|----------------------|---------------------------------------------|---------------------------------------------------------------------------|
| UX-01 | Dashboard dimensions | All four dashboards use the same dimensions | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| UX-02 | Background           | Consistent theme across dashboards          | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| UX-03 | Typography           | Fonts and sizes are consistent              | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| UX-04 | KPI cards            | Consistent design and formatting            | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| UX-05 | Alignment            | Objects are properly aligned                | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| UX-06 | Spacing              | No excessive gaps or overcrowding           | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| UX-07 | Chart labels         | Important values are readable               | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| UX-08 | Legends              | Only useful legends are retained            | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| UX-09 | Filters              | Clearly identifiable and usable             | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| UX-10 | Overall appearance   | Professional and presentation-ready         | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

## 9. Final Delivery QA

| ID    | QA Check                          | Expected Result                           | Status                                                                    |
|-------|-----------------------------------|-------------------------------------------|---------------------------------------------------------------------------|
| FD-01 | Tableau workbook saved            | Final workbook opens successfully         | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| FD-02 | Four dashboards present           | All dashboards are available              | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| FD-03 | Navigation tested                 | All navigation buttons work               | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| FD-04 | Filters tested                    | Filters work as intended                  | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| FD-05 | KPI documentation completed       | All KPIs documented                       | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| FD-06 | Dashboard documentation completed | All four dashboards documented            | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| FD-07 | QA checklist completed            | All required tests recorded               | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| FD-08 | Project folder organized          | Required files are correctly placed       | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| FD-09 | Final presentation checked        | Dashboard screenshots/content are current | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |
| FD-10 | Final submission package checked  | All required deliverables are included    | <input checked="" type="checkbox"/> Pass<br><input type="checkbox"/> Fail |

## 10. QA Summary

| Category          | Total Checks |
|-------------------|--------------|
| Data Quality      | 10           |
| KPI Validation    | 8            |
| Hospital Overview | 10           |

|                       |    |
|-----------------------|----|
| Patient Flow          | 8  |
| Department Analytics  | 8  |
| Resource Analysis     | 9  |
| Dashboard Integration | 10 |
| Visual & Usability    | 10 |
| Final Delivery        | 10 |
| TOTAL                 | 83 |

## 11. QA Notes

- KPI values must be validated against the actual source/final analytical datasets before being marked Pass.
- The Department Efficiency Score must be checked against the exact KPI-generation implementation.
- Dashboard navigation must be tested after all four dashboards are integrated.
- The QA document records verification status and should not claim successful testing before the corresponding check is actually performed.